

THERMOSPHERE

The thermosphere extends to over 375 miles (600 km) above the Earth's surface. Temperature rises rapidly in the lower thermosphere due to absorption of solar energy and then increases gradually with altitude, reaching as high as 3,100°F (1,700°C).

MESOSPHERE

This layer extends up to about 50 miles (80 km). Temperatures fall through the mesosphere to as low as -135°F (-93°C).

STRATOSPHERE

The stratosphere is a calm layer stretching up to about 30 miles (50 km) above sea level. The temperature rises to 27°F (-3°C) at the top of this layer.

TROPOSPHERE

This layer extends to 5 miles (8 km) above the poles and 10 miles (16 km) above the equator. It contains 75 percent of the total mass of the atmosphere. Temperatures fall to as low as -62°F (-52°C) at the top.

HEIGHT ABOVE SEA LEVEL

81 miles
130 km
75 miles
120 km
68 miles
110 km
62 miles
100 km
56 miles
90 km
50 miles
80 km
43 miles
70 km
27 miles
60 km
31 miles
50 km
25 miles
40 km
19 miles
30 km
12 miles
20 km
6 miles
10 km
sea level

aurora

meteor burning up in the atmosphere

ice crystals on meteoric dust

ozone layer absorbs harmful radiation from the Sun

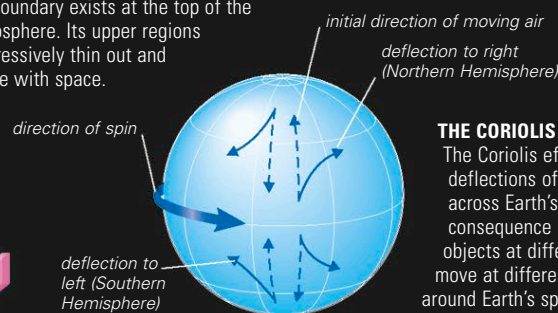
all weather occurs in the lowest level of the atmosphere

ATMOSPHERE AND WEATHER

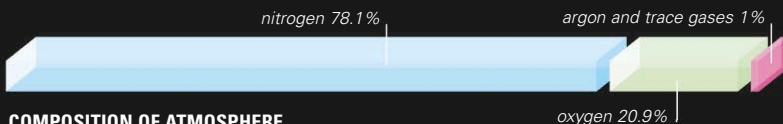
Earth is surrounded by the atmosphere, a layer of gases many hundreds of miles thick. This atmosphere is thought to have arisen partly from gases spewed out by ancient volcanoes, although its oxygen content—so vital to most forms of life—was created mainly by plants. Through the effects of gravity, the atmosphere is densest at Earth's surface and rapidly thins with altitude. With increasing altitude, there are also changes in temperature and a progressive drop in atmospheric pressure. For example, at a height of 19 miles (30 km), the pressure is just 1 percent that at sea level. Within the lowest layer of the atmosphere, the troposphere, continual changes occur in temperature, air flow (wind), humidity, and precipitation, known as weather. The basic cause of weather is the fact that Earth absorbs more of the Sun's heat at the equator than the poles. This produces variations in atmospheric pressure, which create wind systems. The winds drive ocean currents and cause masses of air with different temperatures and moisture content to circulate over the planet's surface. Earth's rotation plays a part in causing this atmospheric circulation because of the Coriolis effect (below).

ATMOSPHERIC LAYERS

The four main layers in Earth's atmosphere are distinguished by different temperature characteristics. No boundary exists at the top of the atmosphere. Its upper regions progressively thin out and merge with space.

**THE CORIOLIS EFFECT**

The Coriolis effect causes deflections of air moving across Earth's surface. It is a consequence of the fact that objects at different latitudes move at different speeds around Earth's spin axis.

**COMPOSITION OF ATMOSPHERE**

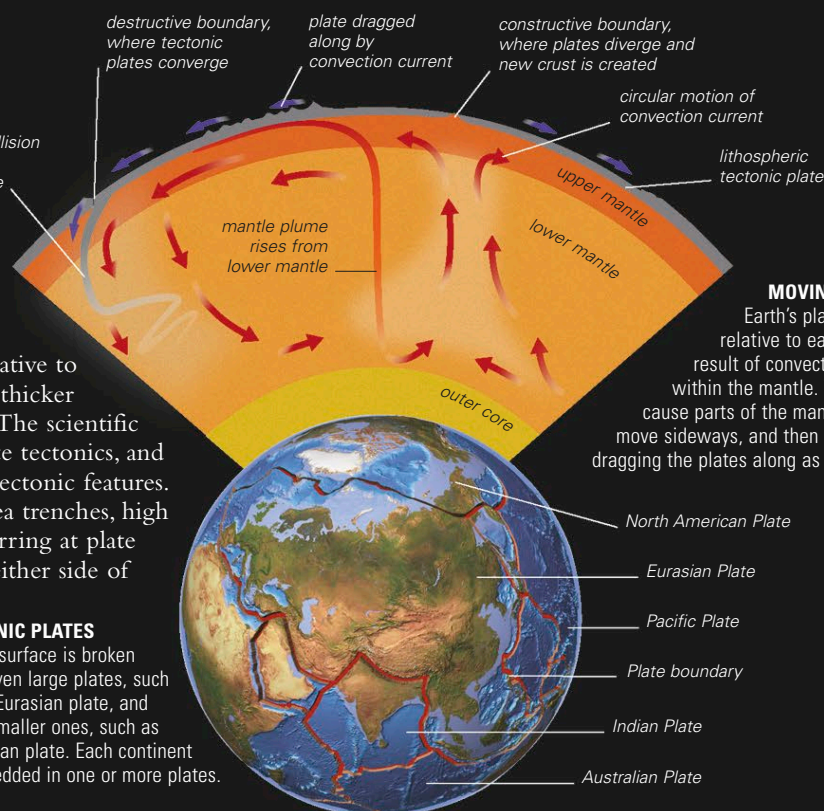
Nitrogen and oxygen make up 99 percent of dry air by volume. About 0.9 percent is argon, and the rest consists of tiny amounts of other gases. The atmosphere also contains variable amounts (up to 4 percent) of water vapor.

PLATE TECTONICS

Earth's crust and the top part of its mantle are joined in a structure called the lithosphere. This is broken up into several solid structures called plates, which "float" on underlying semimolten regions of the mantle and move relative to each other. Most plates carry both oceanic crust and some thicker continental crust, although a few carry only oceanic crust. The scientific theory concerning the motions of these plates is called plate tectonics, and the phenomena associated with the movements are called tectonic features. Most tectonic features, which include ocean ridges, deep sea trenches, high mountain ranges, and volcanoes, result from processes occurring at plate boundaries. Their nature depends on the type of crust on either side of the boundary and whether the plates are moving toward or away from each other. Tectonic features occurring away from plate boundaries include volcanic island chains, such as the Hawaiian islands. These are caused by magma (molten rock) upwelling from "hot spots" in the mantle, causing a series of volcanoes to form on the overlying plate.

TECTONIC PLATES

Earth's surface is broken into seven large plates, such as the Eurasian plate, and many smaller ones, such as the Indian plate. Each continent is embedded in one or more plates.

**MOVING PLATES**

Earth's plates move relative to each other as a result of convection currents within the mantle. The currents cause parts of the mantle to rise, move sideways, and then sink again, dragging the plates along as they do so.